

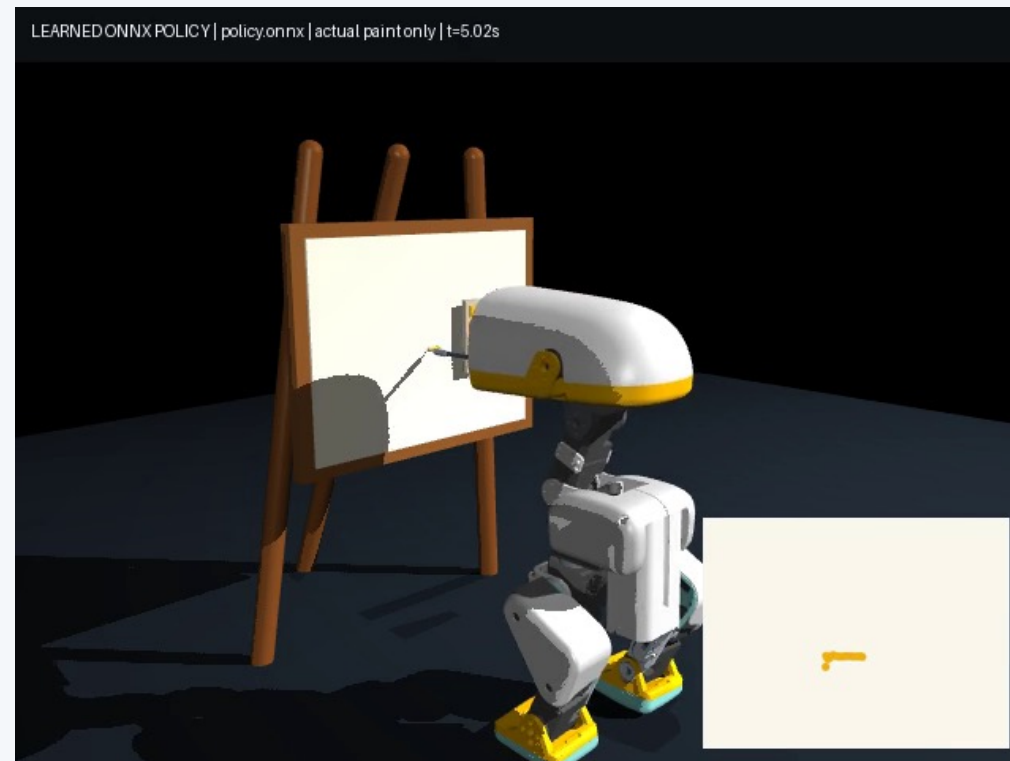
LAB 08

Swimming-duck drawing

Colored-duck painting: successful brush-v2 feedback control

Environment design · PPO · Honest verification

BUILD → TRAIN → VERIFY → EXPLAIN



What you will build — and what the evidence says

- 01** Use case: MicroDuck paints a fixed four-color swimming duck with a preloaded free brush in MuJoCo. The released brush-color-v2-03 ONNX passes 40/40 required episodes. Teach the complete teacher-data, ridge BC, DAgger, anchored PPO, export and verification pipeline. Matching BC also passes 40/40: PPO preserved successful behavior, not demonstrated improvement. Historical pencil failures remain separate evidence.
- 02** By the end: describe the scene, map observations to actions, explain the reward, trace one PPO update, run a bounded experiment, and defend an evaluation claim.
- 03** Successful fixed-template simulated painting is now supported. The earlier pencil experiments were not retroactively successful, and a new 93/15 brush contract is not a replacement for stock 61/14 deployment.
- 04** Scope: local MuJoCo simulation. Do not deploy these classroom artifacts to a physical robot; hardware transfer requires a separate validated process.

Materials and prerequisites • 1/2

- 01** A full-collision MicroDuck simulation, canvas and easel; 14 original joint controls plus one simulated lower-jaw action. This is an educational model, not a hardware bill of materials.
- 02** One free brush held between upper/lower beak pads through the pencil_free joint. It is preloaded, not welded to the head; autonomous pickup is outside this case.
- 03** One passive compliant contact tuft: sphere radius 1.0 mm, mass 0.00003 kg, bristle_compression travel - 3.0 to 0.3 mm. Three decorative hairs and a ferrule are visual, not separate flexible-contact bristles.
- 04** A passive spring with stiffness 25 N/m, damping 0.32 N s/m, armature 0.001 kg and default grip friction 1.2. It adds compliance, not a sixteenth motor.

Materials and prerequisites · 2/2

- 05 Four physical paint wells: gold, orange, charcoal and blue; one non-contact palette backing. Tip contact loads discrete pigment; no fluid mixing, washing or residual-paint dynamics are simulated.
- 06 Existing CPU Python environment with MuJoCo, Torch, Stable-Baselines3, Gymnasium, NumPy, ONNX, ONNX Runtime, Pillow and FFmpeg. Use `rlx/.venv-microduck/bin/python`; source/model compatibility matters more than an unverified package upgrade.
- 07 Retained artifacts: final-seed11 teacher dataset and sidecar; brush-color-v2-03 archived source package, training metadata, BC/PPO checkpoints, frozen ONNX files, evaluations and actual-contact video. A slides-only ZIP does not include every weight or training dependency.

Environment and policy contract · 1/3

- 01** Current contract: microduck-brush-v2, 93 observations and 15 actions, simulation_only with hardware_compatible=false. Historical pencil uses microduck-drawing-v1 83/15. Neither can hot-swap into stock 61/14 walking deployment.
- 02** Observations 0:61 retain IMU angular velocity, projected gravity, 14 relative joint positions, prior joint velocities/actions and zeroed reserved command slots. Observations 61:83 add mouth state, tip/target geometry, velocities, pen-down request and contact forces.
- 03** Brush tail 83:87 = loaded-color one-hot; 87:91 = planned-color one-hot; 91 = compression in millimeters; 92 = trajectory mode divided by 2. The current actor does not consume 83:93: color planning and missed-dip recovery are not learned.
- 04** Actions 0:14 address the original joints; $\text{action}[14]$ maps to $\text{mouth_target} = 0.24 + 0.36 * \text{action}[14]$. The structured head actor predicts a delta bounded to 0.025 per step and adds it to the previous action only at output.

Environment and policy contract · 2/3

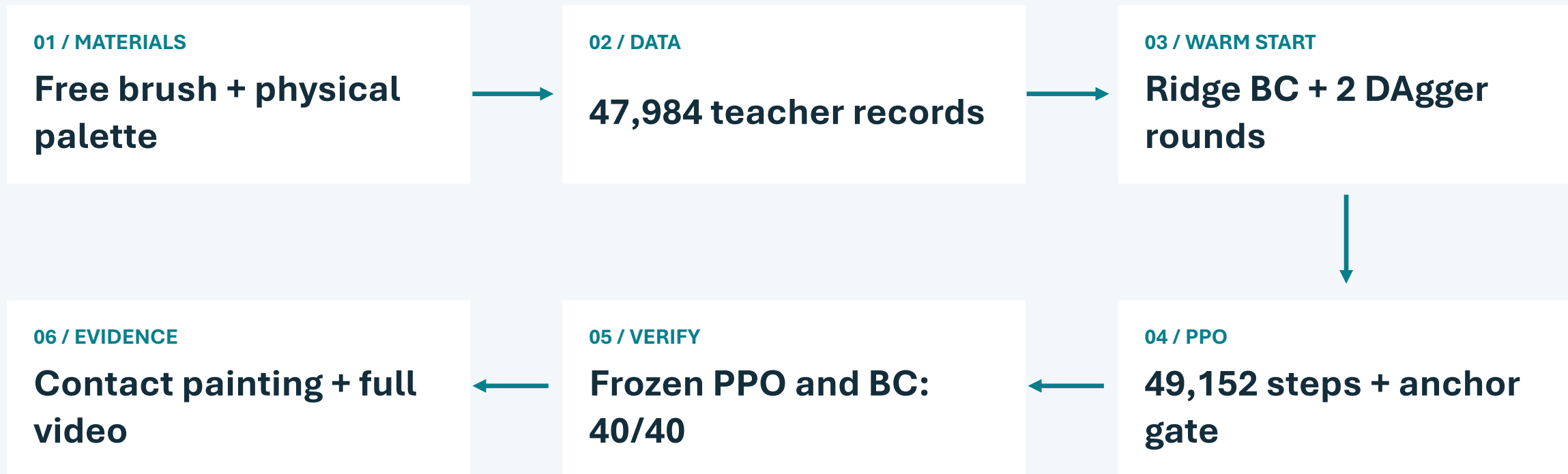
- 05** Head features combine position, target error, brush direction, contact force and quadratic interactions. Ankles use linear gravity/angular-velocity feedback; other body joints and the jaw learn constant offsets. This is a structured learned feedback policy, not an unconstrained image-to-action network.
- 06** Physics timestep 0.002 s gives 500 Hz; 10 substeps give 0.020 s control at 50 Hz. A completed reference trajectory contains 5,998 control steps, measured as 119.96 s despite the 120 s command budget.
- 07** Pigment loads only after actual tip/well force above 0.0001 N. Canvas marks use actual MuJoCo tip contact and the currently loaded color, not reference pixels pasted onto the canvas.
- 08** The fixed duck pattern, color assignment, stroke sequence and approach/dip/lift targets are authored. The learned motor controller follows that plan without a runtime Jacobian teacher. No camera, image recognition or arbitrary visual composition is claimed.

Environment and policy contract · 3/3

09

Success is evaluated independently over ten unique configurations and four seeds per configuration. Reset perturbations test a bounded domain; a single attractive render is not the 40-episode acceptance result.

How this case is built: one evidence loop



Failure is data: change one cause, retrain, and repeat the same evaluation gates.

Build the case step by step · 1/2

- 01** First preserve the history: drawing-pilot-01 and drawing-refinement-02 each used 32,768 PPO transitions and failed pencil acceptance. Their teacher feasibility video is not a successful learned policy.
- 02** Diagnosis found a confirmed configuration defect: intended `initial_std=0.005` was reset to effective 0.02 after BC. Both historical PPO policies also drifted from BC. These observations do not prove a single cause for every failure.
- 03** Build a separate brush-v2 contract with compliant contact, physical color wells and a structured action-delta actor. Multiple changes occurred together, so do not attribute the entire improvement to PPO or brush compliance alone.
- 04** Archive training sources before collecting data. With data seed 11, the analytic Jacobian teacher provides eight 5,998-step episodes: 47,984 initial labeled observations/actions.

Build the case step by step · 2/2

- 05 Fit ridge BC with ridge=0.01. Relabel student-visited states in two DAgger rounds and refit; dataset sizes become 47,984, 77,974 and 105,858. These are supervised fit records, not fresh PPO transitions.
- 06 Warm-start PPO with training seed 101, small per-action noise and learning_rate=1e-7. Collect 49,152 genuine on-policy transitions; evaluate deterministic action drift on saved BC/Dagger anchor observations after each update.
- 07 Accept an update only when maximum drift is finite and ≤ 0.00025 ; otherwise restore policy parameters and optimizer state. Export immutable deterministic PPO and matching BC ONNX with static 93-input/15-output interfaces.
- 08 Run identical independent acceptance suites on both models, verify source/model hashes, inspect all failures and negative controls, then render the same PPO ONNX. The final gate passes; this lesson does not rerun training.

PPO: improve the actor without trusting a huge step

$$\text{ratio} = \pi_{\text{new}}(a \mid o) / \pi_{\text{old}}(a \mid o)$$

Maximize $\text{mean}[\min(\text{ratio} \times A, \text{clip}(\text{ratio}, 1 - \epsilon, 1 + \epsilon) \times A)]$

Actor: chooses joint actions. Critic: estimates future discounted reward.

A = advantage. ϵ = clipping range. Old log probabilities stay fixed for each collected batch.

Follow one rollout through the learner

- 01** Collect observations, sampled actions, old log probabilities, values, rewards and episode-boundary flags using the current stochastic actor.
- 02** Compute TD residuals and generalized advantage estimates (GAE). A true terminal has zero continuation value; a timeout may bootstrap the saved final observation.
- 03** Cut advantage traces at reset boundaries. Never bootstrap a timeout from the first observation of the next episode.
- 04** Reuse this batch for PPO minibatch updates, then collect fresh on-policy data. Export the deterministic actor and its observation transform for evaluation.

Paper exercise: count samples and clip the objective

- 01 Illustrative arithmetic, NOT this run: $16 \text{ environments} \times 128 \text{ steps} = 2,048$ fresh transitions per rollout. Four epochs reuse those transitions; they do not create 8,192 new samples.
- 02 With $\epsilon = 0.2$, ratio = 1.4 and $A = +2$: $\min(2.8, 2.4) = 2.4$. The minimized policy term is -2.4 before averaging and additional loss terms.
- 03 With the same ratio and $A = -2$: $\min(-2.8, -2.4) = -2.8$. Explain why clipping is asymmetric once the sign of the advantage changes.
- 04 Run from repo root: `python3 docs/microduck-training-student-book/learning_lab.py --gae`. This is a deterministic arithmetic lab, not new policy training.

Reward design: what the agent is paid to do · 1/2

- 01 tracking = $2.0 * \exp(-(\text{tip_target_distance} / 0.006)^2)$: a smooth local tracking reward. The 6 mm reward scale is not the stricter 1.5 mm final scoring tolerance.
- 02 contact = 0.4 when actual mark presence matches the requested pen-down state. Correct pen-up travel can also receive this reward; constant contact is not the goal.
- 03 grasp = 0.3 for maintaining grasp through the physics substeps; upright = $0.3 * \max(0, \text{trunk_upright})$. Balance and retention support, but do not replace, image-geometry acceptance.
- 04 action_rate_penalty = $-0.01 * \sum((\text{action} - \text{previous_action})^2)$ discourages abrupt control changes.

Reward design: what the agent is paid to do · 2/2

05 $\text{force_penalty} = -0.02 * \max(0, \text{canvas_force} - 0.5)^2$. Final validation separately enforces peak force below 0.5 N.

06 $\text{color} = 0.4$ when loaded_color equals desired_color . It rewards the achieved contact state; it does not prove the actor learned a color plan, because the actor ignores the added color tail.

07 The environment returns the sum of these step rewards. Logs retain reward_mean , not a complete episodic-return series. A higher reward or accepted anchor update does not automatically mean a better painting.

Recorded training recipe · 1/3

- 01 Selected run: `rlx/runs/studio/drawing/brush-color-v2-03`. Data seed 11; training seed 101; dataset SHA-256 is recorded in `training.json` and validated against the teacher sidecar before fitting.
- 02 Ridge BC: 0.01; two DAgger rounds. Normalized increment MSE after fits: $8.067842827585991e-06$, $1.9331184375914745e-05$ and $3.993013524450362e-05$. Changing datasets make these values unsuitable as a single monotonically comparable learning curve.
- 03 One environment; `n_steps=6144`; eight rollouts produce 49,152 fresh transitions. `batch_size=512` gives 12 minibatches per epoch; `n_epochs=2` reuses data rather than doubling simulator transitions.
- 04 PPO `learning_rate=1e-7`; `gamma=0.995`; `gae_lambda=0.95`; `clip_range=0.05`; `target_kl=0.002`; `max_grad_norm=0.3`. These values are from the archived training script, not generic PPO defaults.

Recorded training recipe · 2/3

- 05 Value coefficient `vf_coef=0.05`; entropy coefficient `ent_coef=0`. Exploration standard deviations: head 0.0003, body/legs 0.0001, jaw 0.0002. The logged `entropy_loss` is not the entropy itself.
- 06 Anchor observations are subsampled from BC/DAGger data using a stride based on 4,096. Each update compares deterministic actions against the saved BC/DAGger actor; maximum allowed drift is 0.00025.
- 07 All eight PPO updates passed the drift gate. The largest recorded `anchor_max_drift` is 0.00015974044799804688. Acceptance means the action-drift constraint was met, not a causal gain in image quality.
- 08 ONNX parity `max_absolute_error=1.4901161193847656e-08`, `passed=true`. PPO ONNX SHA-256: 6e083a2a1a0e3114016be9ef0d377d9e9245648ca174303c5f8d0340bdf4500e.

Recorded training recipe · 3/3

09

Accounting exercise: 47,984 initial teacher records; 105,858 final fit records; 49,152 PPO transitions; $40 * 5,998 = 239,920$ evaluation steps per model; 3,484 contact rows in the showcase trace. These are different data populations.

Hands-on commands: inspect, smoke, train · 1/3

01

```
From .:  
PYTHONPATH=rlx:microduck_local/src rlx/.venv-microduck/bin/python  
rlx/examples/ppo_microduck_brush.py --help  
Inspect the brush CLI and its separate 93/15 interface before running an experiment.
```

02

```
From .:  
DATA="$(mktemp -d /tmp/microduck-brush-data.XXXXXX)"; PYTHONPATH=rlx:microduck_local/src  
rlx/.venv-microduck/bin/python rlx/examples/ppo_microduck_brush.py data --out "$DATA" --  
seed 11; printf '%s\n' "$DATA"  
Create teacher data in a fresh temporary directory. Retain the printed DATA path for the  
next command; never overwrite the published dataset.
```

Hands-on commands: inspect, smoke, train · 2/3

03

From .:

```
RUN="$(mktemp -d /tmp/microduck-brush-run.XXXXXX)"; PYTHONPATH=rlx:microduck_local/src  
rlx/.venv-microduck/bin/python rlx/examples/ppo_microduck_brush.py train --out "$RUN" --  
data "$DATA/teacher.npz" --seed 101 --steps 49152 --dagger 2 --ridge 0.01 --learning-rate  
1e-7 --anchor-limit 0.00025
```

In the same shell, train ridge BC, two DAgger rounds and anchored PPO into a fresh RUN.
This reproduction is a new experiment, not a guaranteed pass.

04

From .:

```
OUT="$(mktemp -d /tmp/microduck-brush-eval.XXXXXX)"; PYTHONPATH=rlx:microduck_local/src  
rlx/.venv-microduck/bin/python rlx/examples/ppo_microduck_brush.py eval --onnx  
rlx/runs/studio/drawing/brush-color-v2-03/policy.onnx --out "$OUT/ppo-eval.json" --seed  
10001 --episodes 4
```

Evaluate the published PPO ONNX on ten conditions times four seeds; write a separate
report, not over the source evidence.

Hands-on commands: inspect, smoke, train · 3/3

05

From .:

```
OUT="$(mktemp -d /tmp/microduck-brush-bc-eval.XXXXXX)"; PYTHONPATH=rlx:microduck_local/src  
rlx/.venv-microduck/bin/python rlx/examples/ppo_microduck_brush.py eval --onnx  
rlx/runs/studio/drawing/brush-color-v2-03/bc.onnx --out "$OUT/bc-eval.json" --seed 10001 --  
episodes 4
```

Evaluate matching BC with the identical gate; do not compare PPO against an unmatched teacher or different task.

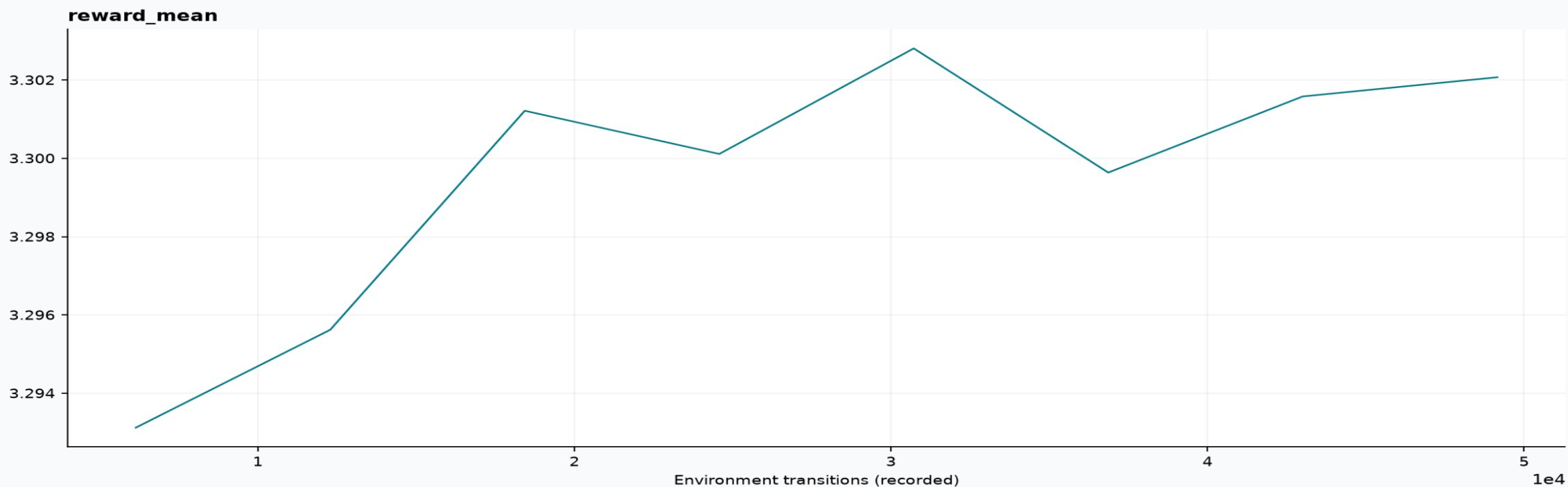
06

From .:

```
OUT="$(mktemp -d /tmp/microduck-brush-render.XXXXXX)"; PYTHONPATH=rlx:microduck_local/src  
rlx/.venv-microduck/bin/python rlx/examples/render_microduck_brush.py --onnx  
rlx/runs/studio/drawing/brush-color-v2-03/policy.onnx --out "$OUT" --seed 501 --seconds 120  
Render the published learned ONNX with no runtime teacher. The 120 s request completes a  
119.96 s authored trajectory.
```

Reward: does the logged signal match the task?

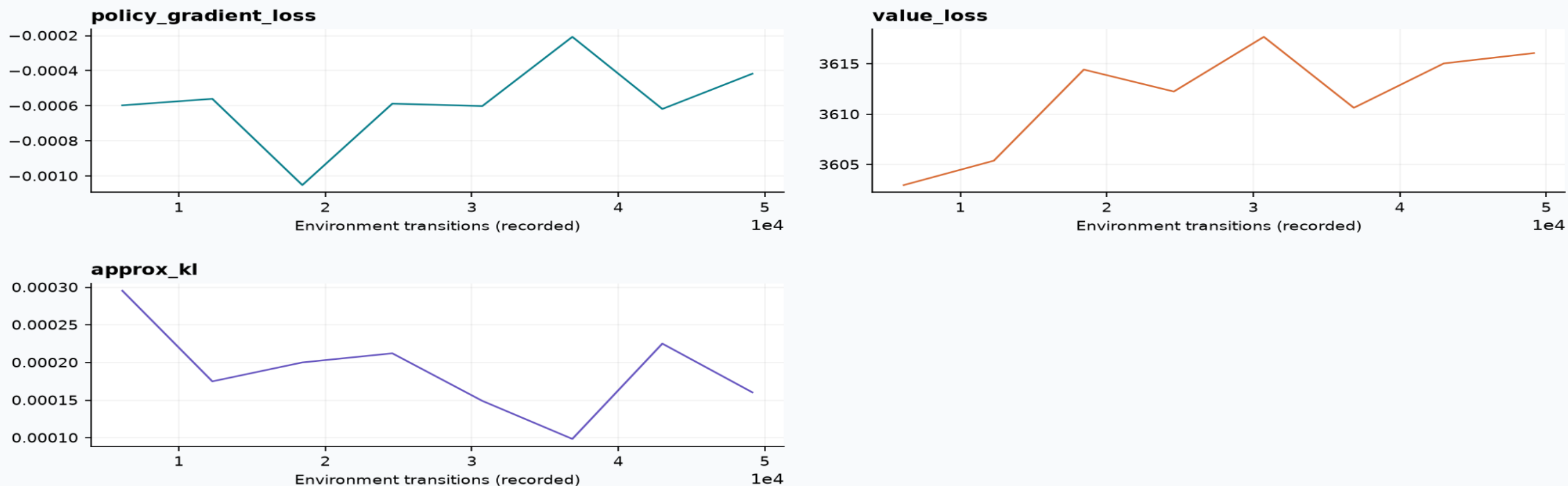
Recorded reward signals — different definitions, not interchangeable



Training reward is an optimization signal, not the acceptance test. See [raw/<case>/](#) for complete recorded fields.

Loss: inspect actor, critic and policy drift

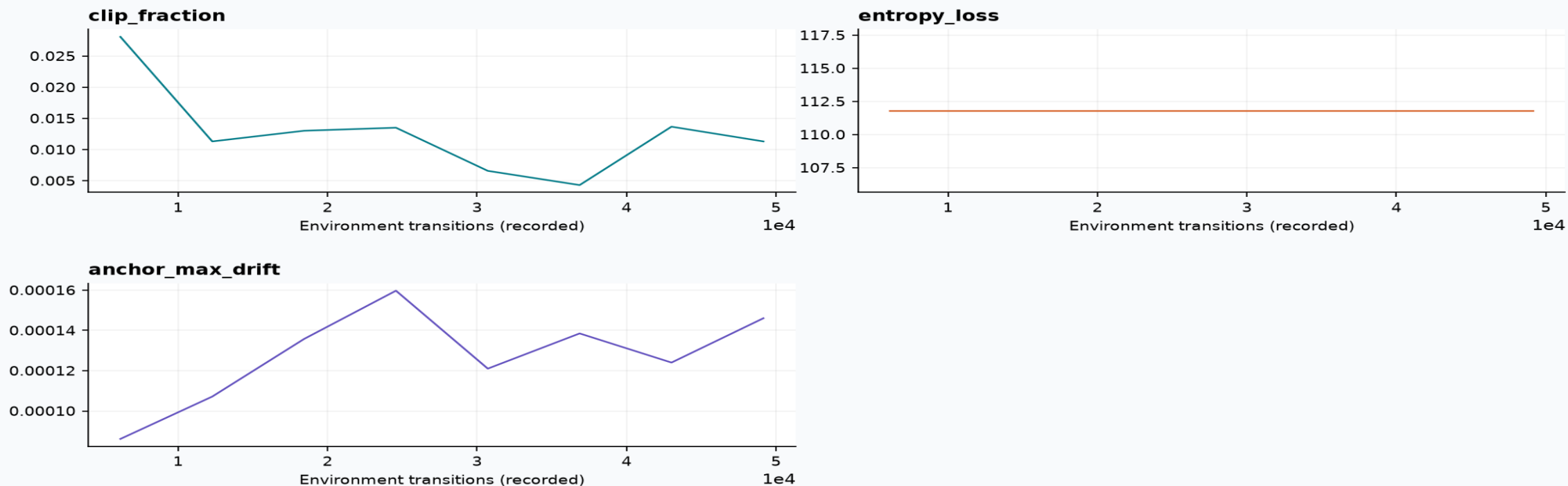
Recorded PPO optimization — raw values, no smoothing



A lower loss does not prove a better robot skill. Loss sign and scale depend on the implementation.

Diagnostics: is the optimizer behaving?

Recorded update diagnostics — absent fields are explicit



Entropy, clipping, explained variance and timing are diagnostics — not a standalone pass/fail certificate.

Metric matrix: numerical audit of this trace · 1/2

- 01 anchor_max_drift: first 0.00008613 → final 0.0001460; min 0.00008613, max 0.0001597; 8 logged samples.
- 02 approx_kl: first 0.0002954 → final 0.0001603; min 0.00009847, max 0.0002954; 8 logged samples.
- 03 clip_fraction: first 0.02808 → final 0.01131; min 0.004313, max 0.02808; 8 logged samples.
- 04 entropy_loss: first 111.8 → final 111.8; min 111.8, max 111.8; 8 logged samples.

Metric matrix: numerical audit of this trace · 2/2

05

policy_gradient_loss: first -0.0005970 → final -0.0004170; min -0.001051, max -0.0002070; 8 logged samples.

06

reward_mean: first 3.293 → final 3.302; min 3.293, max 3.303; 8 logged samples.

07

value_loss: first 3603 → final 3616; min 3603, max 3618; 8 logged samples.

Verification: define success before looking · 1/3

- 01** Freeze model bytes before evaluation. Check ONNX, checkpoint, sidecar, embedded training metadata and archived training-source hashes; inspect provenance_errors rather than treating a filename as provenance.
- 02** Ten required configurations: nominal; offset (0.002,0.002) m; scale 0.95; grip friction 0.9; brush stiffness 30 N/m; combined_left; combined_small; combined_right; boundary_large; boundary_soft. Each must pass all four seeds.
- 03** Combined gates vary offset, scale, friction and stiffness together. Boundaries reach scale 0.88 to 1.12, grip friction 0.8 and stiffness 18 to 40 N/m; exact configurations are retained in raw/drawing/eval.json and the source workflow.
- 04** For each of gold, orange, charcoal and blue: coverage ≥ 0.95 and precision ≥ 0.95 at geometric tolerance 0.0015 m. Require a full 5,998-step trajectory, no fall/drop and assistance=0.

Verification: define success before looking · 2/3

- 05 Safety-style simulation gates: `grasp_fraction >=0.90`, `max_normal_force_n <0.5`, `max_compression_m <0.0035`, `pen_up_leak_fraction <0.05`. These are simulation acceptance checks, not physical-robot safety certification.
- 06 Require actual physical palette loading in gold -> orange -> charcoal -> blue order. Both null and open_jaw negative controls must fail. Reference-target substitution or merely showing a colorful texture is not valid evidence.
- 07 Evaluate matching BC with the same condition and seed suite. PPO 40/40 and BC 40/40 support preservation; neither success rate nor the slightly lower PPO precision demonstrates PPO superiority.
- 08 Both final PPO and matching BC reports allow an evaluator-only pipeline change: `training_pipeline_source_match=false`, `environment_source_match=true` and `provenance_errors=[]`. The archived training script still matches each model. Matching BC hardened re-evaluation completed on September 11, 2026 at 21:41:35 UTC.

Verification: define success before looking · 3/3

09

Bind render source_sha256 to the evaluated PPO ONNX. Separately inspect the browser receipt: viewer.json reports passed=false for a bounded live check. A successful offline ONNX suite does not imply every UI verification passed.

Results: what was actually measured • 1/3

- 01 Released brush-color-v2-03: passed=true, skill_status=passed, 40/40 required PPO episodes, ten unique configurations and four seeds each. Each run completes 5,998 steps / 119.96 s with assistance=0.
- 02 PPO aggregate mean_coverage=1.0 and mean_precision=0.999531977503557. Matching BC also passes 40/40, with mean_coverage=1.0 and mean_precision=0.9996346455645799. Conclusion: PPO preserved successful BC/Dagger behavior.
- 03 PPO worst per-color precision is 0.9862436754827598, above 0.95; minimum grasp_fraction is 0.9998332777592531. Aggregate coverage must not hide a failed individual color or seed.
- 04 PPO peak force 0.040318330721211285 N, peak compression 0.0015517988613261476 m and maximum pen-up leak 0.04143426294820717 all satisfy the declared gates. These are maxima over the required evaluation suite, not a single frame.

Results: what was actually measured · 2/3

- 05 All eight recorded updates passed the anchor gate. reward_mean moves from 3.2931220531463623 to 3.3020718097686768; the small change is not proof of improved painting.
- 06 Final policy_gradient_loss=-0.0004169998671083401; value_loss=3616.0631205240884; approx_kl=0.0001603432756382972; clip_fraction=0.011311848958333334. Value loss remains large despite accepted skill behavior: optimizer diagnostics and task success differ.
- 07 Source-bound showcase: controller=onnx, teacher_diagnostic=false, assessment passed=true, 3,484 recorded contact rows and all four colors. The actual painted image comes from the contact trace, not the authored reference image.
- 08 History remains intact: the two pencil PPO trials failed; brush-color-v2-01 was preliminary 20/20; brush-color-v2-02 failed BC validation with data seed 101; final brush-color-v2-03 uses data seed 11 and train seed 101.

Results: what was actually measured · 3/3

09

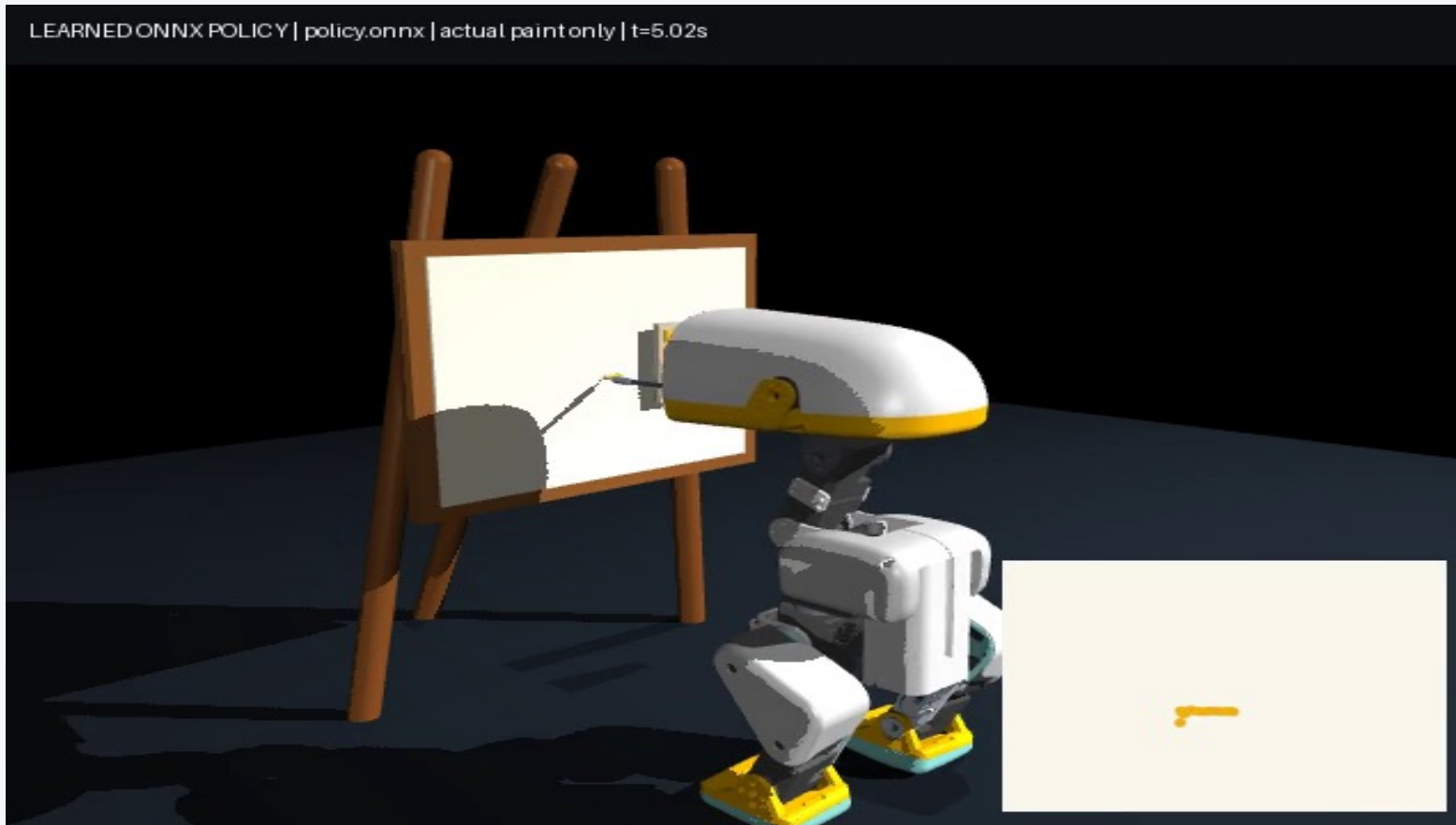
Independent boundaries: final PPO and matching BC both report `provenance_errors=[]` and `environment_source_match=true`. The browser/live receipt still reports `passed=false`. Do not silently merge independent model acceptance and browser verification.

Successful rollout: actual contact painting



Actual four-color brush contact painting from the released PPO ONNX; not the reference target. The full 40-episode suite passes, and matching BC also passes.
Source: docs/drawing-case/brush/media/render.json.

Watch the actual saved rollout



BEFORE PLAY

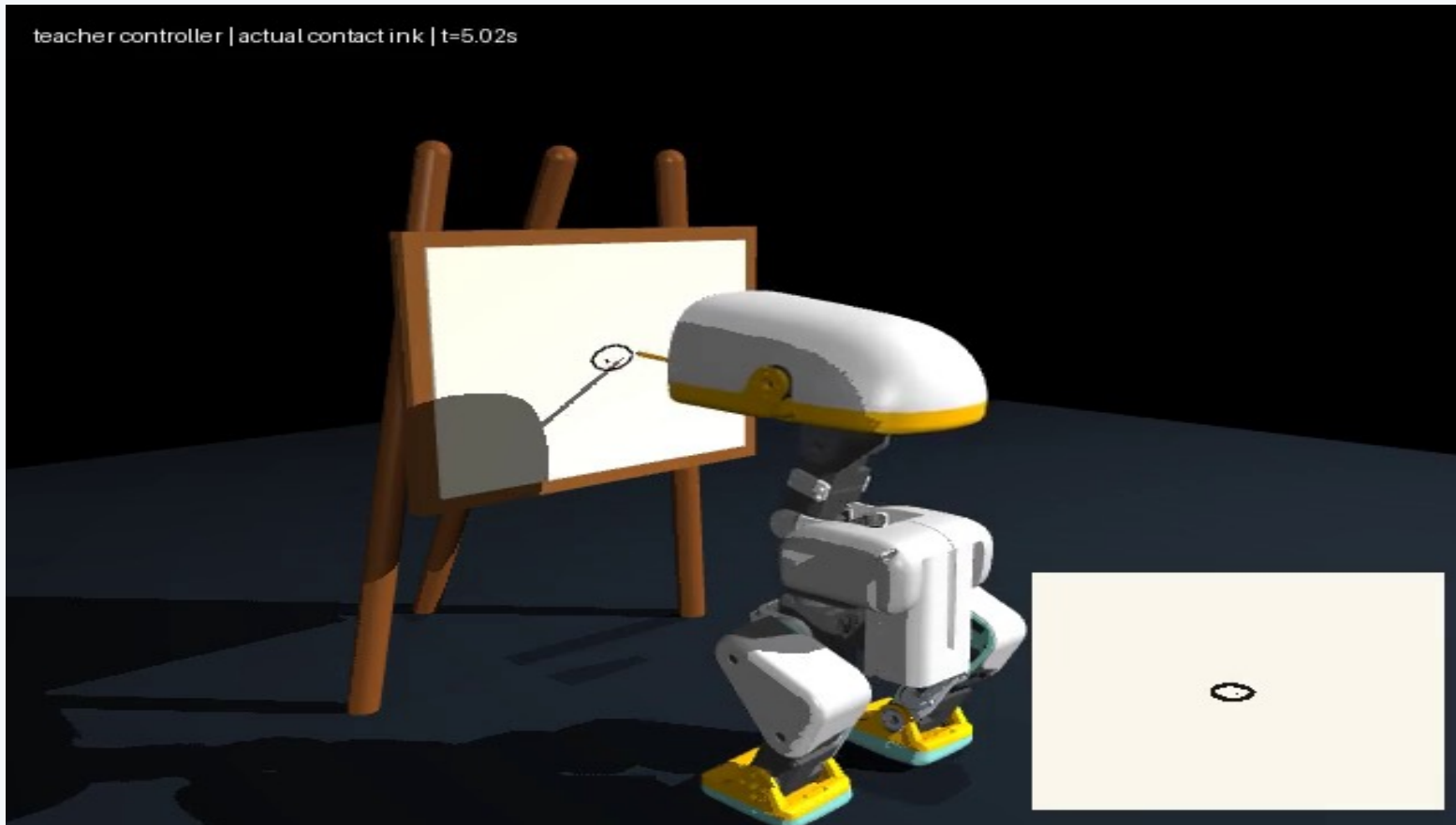
Predict one success signal and one failure signal.

AFTER PLAY

Which claims does this clip support? What remains untested?

120.0 s saved MP4 • Click in PowerPoint Slide Show • Backup: videos/drawing-rollout.mp4

Historical pencil teacher: a different 83/15 experiment



BEFORE PLAY

Predict one success signal and one failure signal.

AFTER PLAY

Which claims does this clip support? What remains untested?

Historical pencil feasibility only. This is an analytic teacher, not the brush-v2 learned PPO policy, not the matching BC baseline, and not evidence that the old pencil PPO trials passed. • 32.0 s saved MP4 • Click in PowerPoint Slide Show • Backup: videos/drawing-teacher.mp4

Limits and failure analysis · 1/2

- 01 Successful fixed-template simulated painting is now supported. The earlier pencil experiments were not retroactively successful, and a new 93/15 brush contract is not a replacement for stock 61/14 deployment.
- 02 BC/Dagger already passed the same 40/40 benchmark with slightly higher precision. No PPO improvement, from-scratch acquisition or single-factor causal explanation is established.
- 03 The actor ignores color-tail observations 83:93; color and dip order are authored. Do not describe the system as learned color planning, vision, arbitrary drawing or autonomous missed-dip recovery.
- 04 The brush starts in the mouth. Pickup, washing, pigment mixing, material calibration, real servo safety and hardware transfer remain untested. Decorative hairs are not independently simulated flexible bristles.

Limits and failure analysis · 2/2

05

The failed brush data-seed-101 attempt demonstrates seed sensitivity. Repeating the recipe with a different seed is an experiment, not a promised classroom pass.

06

Anchor acceptance limits action drift only on selected training observations. It is not an independent generalization or painting-quality certificate; the 40-episode ONNX suite provides separate evidence.

07

This classroom update reuses existing training/evaluation artifacts and video; it performs no new training. Browser verification and any future model re-evaluation must be tracked independently.

Student lab: change one variable, keep the gates · 1/2

- 01 Explain the three contracts: stock 61/14, historical pencil 83/15 and current brush 93/15. Identify the passive compliance coordinate that is not an additional action.
- 02 Compute fresh PPO transitions as $1 * 6144 * 8 = 49,152$. Explain why two optimizer epochs, teacher records and contact rows are not extra on-policy simulator transitions.
- 03 Trace one approach -> dip -> paint sequence. Distinguish the authored target plan, learned motor action, physical pigment loading and contact-based scoring.
- 04 Read `reward_mean`, `policy_gradient_loss`, `value_loss`, `approx_kl`, `clip_fraction`, `entropy_loss` and `anchor_max_drift` from the eight-row CSV. Explain why `entropy_loss` is not entropy and why `accepted_update` is not a success-rate curve.

Student lab: change one variable, keep the gates · 2/2

05

Propose one controlled change: anchor limit, exploration noise, or training-data seed. Keep the BC baseline, source hashes, evaluation conditions and acceptance thresholds fixed; retain rejected updates and failed runs.

06

Compare the 40/40 BC and PPO tables. Write the strongest justified claim, then name the missing evidence needed to claim PPO improved quality, generalized to unseen artwork, or worked on a physical robot.

Submit an experiment, not just a video

- 01** Before training: write one hypothesis, the changed parameter, the fixed baseline, seed list, acceptance criteria and time/transition budget.
- 02** During training: save the exact command, environment versions, initialization, checkpoint and raw reward/loss/KL metrics. Never overwrite the saved reference run.
- 03** After training: evaluate the frozen exported policy on matched seeds; include all failures, aggregate metrics, the full rollout video and a source-bound result table.
- 04** Rubric / 10: reproducibility 2; environment and reward explanation 2; PPO and metrics interpretation 2; honest verification 2; one evidence-based next experiment 2.

Source map and further study · 1/2

01

`docs/drawing-case/brush/README.en.md`

02

`docs/drawing-case/brush/RESULTS.md`

03

`docs/drawing-case/brush/DIAGNOSIS.md`

04

`docs/drawing-case/brush/training.json`

Source map and further study · 2/2

05

`docs/microduck-training-student-book/chapters/02a-ppo-code-lab.md`

06

`docs/microduck-training-student-book/chapters/02b-gae-workshop.md`

07

`scripts/scenarios/README.md`

08

`README.md`